TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 2: Ethics in Engineering

Why Ethics Belongs in Engineering, Professional Responsibility, Fukushima Case, Code of Ethics & Academic Integrity

PART 2

Ethics in engineering

3 sections, about 1 h 30 min of reading.

Why ethics belongs in engineering

~25 min to read | Medium priority | 3 questions

Source: Set 2, slides 1-24

What is meant by ethics, and the first clicker question

The lecture opens with a plain definition. **Ethics** is a *system of moral principles*. It is the set of *principles of right and wrong*. It is also the *principles of conduct governing the behaviour of an individual or a group*. Three phrases to remember: system of moral principles, principles of right and wrong, principles of conduct governing behaviour of an individual or a group. For the full treatment of ethics, morals, values and law, and the difference between them, go to section t01.

Clicker question 1. A person's behaviour is always ethical when one: A. does what is best for oneself; B. has good intentions, no matter how things turn out; C. does what is best for everyone; D. does what is legal.

- A is ethical egoism. It ignores everyone else, so it cannot make behaviour *always* ethical.
- B judges only the intention and ignores the outcome. This is the duty-based (deontological) view taken to an extreme.
- C is the consequentialist, utilitarian idea of the greatest good for everyone affected. It is the closest to the lecture's way of thinking, because the lecture keeps stressing consequences for society.
- D is wrong because law and morality are different things. An act can be legal and immoral. The lecture returns to this later (see section t06).

beyond the slides The slide does not print an answer. C is the usual intended answer, but no single one-line test makes behaviour "always" ethical. Treat the question as a warning: the other options are the traps (selfishness, intention only, legality only).

Where does ethics fit into an engineering course?

The lecturer asks the awkward question directly. Students have been studying design, analysis and performance measurement. So why is there a course on ethics? The answer comes in the next slides and it is the heart of this section: engineers build things that act on society, so their decisions carry moral weight.

Engineers build products. The slide lists cell phones, home appliances, heart valves, bridges and cars. In general, engineers advance society by building new technology.

Engineers develop processes. The examples are the process that converts salt water into fresh water and the process that recycles bottles. These processes change how we live and what we can accomplish.

Products and processes have consequences for society

The lecture gives four short "if" examples. Learn them, because they are the quickest evidence you can quote in an essay.

- If a bridge has inadequate support, it will fail.
- If a gas tank is positioned too close to the bumper, it might explode from a small accident.
- If a medical instrument is not accurate, improper doses of medication can be given.
- If a process for refining gas produces too many toxins, it harms the local community.

The pattern is the same each time. A technical choice leads to a consequence for people. The next slide states the principle: *decisions made by engineers usually have serious consequences to people, often to multitudes of people*. Therefore *ethics and ethical reasoning guide decision-making*.

> **Key point.** The argument in one chain: engineers make products and processes; these have consequences for society, often for many people; so engineers' decisions need ethical reasoning to guide them.

MARKER WORDS: WHAT ETHICS IS AND WHY CONSEQUENCES MATTER

Say it like this:

> Ethics is a **system of moral principles**: the **principles of right and wrong** that govern the **conduct of an individual or a group**. Engineers' products and processes have **serious consequences** for **society**, so **ethical reasoning guides decision-making**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
consequences for society	explaining why engineering is not value-neutral	SHOULD
society	saying who is affected by an engineer's work	MUST
principles of **right and wrong**	defining ethics	SHOULD
multitudes of people	stressing scale of impact	SHOULD
products and **processes**	listing what engineers produce	SHOULD
ethical reasoning	saying what guides decisions	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is ethics (lecture wording)?
A system of moral principles; principles of right and wrong; principles of conduct governing the behaviour of an individual or a group.

The four "if" examples of consequences of engineering products and processes.
Bridge with inadequate support fails; gas tank too close to the bumper may explode from a small accident; inaccurate medical instrument gives improper doses; gas-refining process with too many toxins harms the local community.

What guides decision-making when engineers' decisions have serious consequences for people?
Ethics and ethical reasoning.

Worked example: Fukushima and the ethics of nuclear power

The lecture uses one case to show ethical reasoning at work. On 11 March 2011 an earthquake of magnitude 8.9 struck near Sendai, Japan. It damaged the Fukushima I Nuclear Power Plant (Fukushima Dai-ichi). The damage led people worldwide to rethink the ethics of nuclear power. The slides then list the issues that come up in such discussions. Notice that the issues are not only about the reactor. They also cover the *alternatives* to nuclear power.

Issue 1: health and safety risks

The danger is to current and future generations from leakage of the radio-isotopes used in nuclear power. Plutonium-239 has a half-life of 24,110 years and is a particularly toxic radio-isotope. Normally, 10 half-lives must pass before a Pu-239 contaminated area is considered safe again. For plutonium that is roughly 250,000 years (10 x 24,110 = 241,100, rounded up). If plutonium leaked, say because of an earthquake, it would be a health risk for roughly 8000 generations. (The slide's figure implies about 30 years per generation.)

Two further considerations complicate this issue:

- **(a) Uncertainty.** Medical science may discover a cure for cancer in the current or next centuries. That adds uncertainty to the long-term health risks of leakage.
- **(b) Possible benefit.** Using nuclear power may increase our knowledge of the radio-isotopes used for medical purposes.

Issue 2: depletion of resources

This issue concerns the consequences of the alternatives to nuclear power. Fossil fuels (oil, natural gas and coal) are non-renewable. They also work against the goal of health because of their effect on pollution and climate change.

Issue 3: comparative economic costs of renewable sources

Renewable sources are hydro-electric power, wind power, solar power, geothermal heat, agricultural biomass and tides. They do not cause the environmental hazards that fossil fuels do. But they must be balanced against the energy needed to produce and maintain them, and against the environmental hazards that follow from that. The slide's example: at present, the energy needed to manufacture and install solar energy systems comes from fossil fuels.

The reasoning behind the issues: goals and means

The slide on reasoning explains what kind of thinking is going on. It involves certain **goals**: in this case health, safety and biodiversity. The reasoning then looks for the best, or at least a reasonably better, **means** of obtaining those goals. This type of reasoning is called **practical reasoning**. It uses different methods from mathematics and the sciences. **Ethical reasoning** is a type of practical reasoning that concerns certain societal or life-form goals, such as justice, equality, freedom, health and safety.

So the Fukushima discussion is a worked example of practical reasoning. Set it out as a table: fix the goals, list the possible means, test each means against the goals.

Means (energy source)	Helps the goals	Threatens the goals
Nuclear power	Does not burn fossil fuels; may add medical knowledge about radio-isotopes	Leakage of Pu-239 is toxic for about 250,000 years (about 8000 generations); a cure for cancer is uncertain
Fossil fuels (oil, gas, coal)	Existing supply of energy	Non-renewable, so resources are depleted; pollution and climate change harm health
Renewables (hydro, wind, solar, geothermal, biomass, tides)	No fossil-fuel type environmental hazards	Energy and cost to build and maintain; solar systems are still made with fossil-fuel energy

> **Key point.** No means is perfect. Practical reasoning does not prove one answer the way a mathematical proof does. It weighs the means against the goals and picks the best, or a reasonably better one. That weighing is what the lecturer calls ethical reasoning when the goals are societal ones such as health and safety.

The same example shows how the three issues connect. Issue 1 is about the risk of one means. Issues 2 and 3 are about the consequences of the alternatives. A good ethical argument looks at all of them, not at the reactor alone. For the wider environment material (pollution, fossil fuels, sustainable development) see section t22.

MARKER WORDS: PRACTICAL REASONING AND ETHICAL REASONING

Say it like this:

> **Practical reasoning** sets **goals**, such as health, safety and biodiversity, and then finds the best or a reasonably better **means** of obtaining them. **Ethical reasoning** is a type of practical reasoning concerned with **societal goals** such as justice, equality, freedom, health and safety.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
health and safety risks	describing the danger of a technology to people	SHOULD
pollution and climate change	describing the cost of fossil fuels	SHOULD
uncertainty	saying the long-term effects are not known for sure	MUST
current and future generations	stressing how long a risk lasts	SHOULD
depletion of resources (non-renewable)	criticising fossil fuels	SHOULD
comparative economic costs of renewables	weighing alternatives fairly	SHOULD
goals and **means**	explaining practical reasoning	SHOULD
biodiversity	naming an environmental goal	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is practical reasoning?
Reasoning that involves certain goals (here health, safety and biodiversity) and focuses on finding the best, or at least a reasonably better, means for obtaining them. It uses different methods from mathematics and the sciences.

What is ethical reasoning?
A type of practical reasoning that concerns certain societal or life-form goals, such as justice, equality, freedom, health and safety.

The three issues in the Fukushima discussion.
Issue 1: health and safety risks. Issue 2: depletion of resources. Issue 3: comparative economic costs of renewable sources.

The six renewable sources listed in the slides.
Hydro-electric power; wind power; solar power; geothermal heat; agricultural biomass; tides.

The three fossil fuels named in the slides.
Oil, natural gas and coal (non-renewable).

The essence of an engineering career, and social responsibility

The lecture now states why this matters to *your* career. Engineering is one of the most important professions in society. As engineers we do not just build things and develop processes. We build things and make processes *in order to better society*. To make society better we have to *reflect constantly* on the products and processes that we make.

Social responsibility is the first main connection between ethics and engineering. It comes from the impact that engineered products and processes have on society. Engineers have to think about designing, building and marketing products that benefit society. Social responsibility requires taking into consideration the needs of society.

Typical ethical issues that engineers encounter

Learn this list whole. The slide gives eleven items, the last being open-ended:

- Safety
- Acceptable risk
- Compliance
- Confidentiality
- Environmental health
- Data integrity
- Conflict of interest
- Honesty/dishonesty
- Societal impact
- Fairness
- Accounting for uncertainty, etc.

Link them to cases: safety and acceptable risk are the subject of section t16; conflict of interest is in section t21.

Professional responsibility and the two dimensions of ethics in engineering

Ethics has a **second** connection with engineering. Being socially responsible puts duties and obligations on us *individually*. Ethics therefore also fits into engineering through **professional responsibility**.

Put the two connections together. The slide "Two dimensions of ethics in engineering" says ethics is part of engineering for two main reasons:

1. Engineers need to be **socially responsible** when building products and processes for society.
2. **Social responsibility requires professional responsibility.**

In short: the first dimension is about what engineering does to society; the second is about the personal duties that fall on the engineer because of it.

What ABET says

ABET is the US body that accredits engineering programmes. The slide quotes the outcome it requires: by graduation, students will have *an understanding of professional and ethical responsibility*. So ethics is not an add-on to an engineering degree. It is part of what an accredited degree must produce. **beyond the slides** ABET is the Accreditation Board for Engineering and Technology. Your own university's accreditors set similar outcomes, which is why this course is examined.

What the course will discuss

The closing slide lists four coming topics: the code of ethics for engineers (section t11); practising ethics as an engineering student (section t05); how to identify and analyse ethical dilemmas through case analysis (t06 and t07); and specific examples of ethical situations you may encounter (t04 and t08). For what makes engineering a profession and what motivates engineers to practise well, see section t21.

MARKER WORDS: SOCIAL AND PROFESSIONAL RESPONSIBILITY

Say it like this:

> Ethics belongs in engineering for **two reasons**. First, engineers must be **socially responsible** because their products and processes have an **impact on society**. Second, **social responsibility requires professional responsibility**, which puts **duties and obligations** on each engineer individually.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
professional responsibility	explaining the personal duty of the engineer	**MUST**
social responsibility	explaining the link from products to society	**MUST**
safety, health and welfare of the public	stating the engineer's first duty	**MUST**
integrity, honesty, fairness, confidentiality	listing typical ethical issues	**MUST**
better society / benefit society	stating the purpose of engineering	**MUST**
needs of society	defining social responsibility	**SHOULD**
reflect constantly on products and processes	describing the engineer's attitude	**SHOULD**
ABET: professional and ethical responsibility	showing ethics is an accreditation outcome	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The two dimensions (reasons) of ethics in engineering.
(a) Engineers need to be socially responsible when building products and processes for society. (b) Social responsibility requires professional responsibility.

Social responsibility (lecture wording).
Taking into consideration the needs of society. It comes from the impact engineered products and processes have on society, so engineers must design, build and market products that benefit society.

Typical ethical issues engineers encounter.
Safety; acceptable risk; compliance; confidentiality; environmental health; data integrity; conflict of interest; honesty/dishonesty; societal impact; fairness; accounting for uncertainty, etc.

What does ABET say about students at graduation?
By the time of graduation students will have an understanding of professional and ethical responsibility.

What the course will discuss (four items).
The code of ethics for engineers; practising ethics as an engineering student; how to identify and analyse ethical dilemmas through case analysis; specific examples of ethical situations you may encounter.

Exam questions on why ethics belongs in engineering

These two questions are the reason this section matters. The first is a 12-mark essay that has appeared in several papers. Note that one paper (QB) states it the other way round: "Ethics has a place in the practice of the engineering profession." Answer it the same way. The strategy is: take a position, give about six reasons, include the opposing view briefly, and rebut it. Motivation for engineers to behave ethically is covered in section t21.

Q: **2011 / 2013 / 2015 papers (also QB, reversed wording) · 12 marks** **"Ethics has no place in the practice of the engineering profession." Discuss this statement.**

Q: **2017 Q3E · 5 marks** **Explain the need for ethics in the engineering profession.**

Q: **Set 2 slide 3-4 clicker · practice** **A person's behaviour is always ethical when one: A. does what is best for oneself; B. has good intentions, no matter how things turn out; C. does what is best for everyone; D. does what is legal.**

The engineering code of ethics and its cases

~40 min to read | High priority | 4 questions

Source: Set 2, slides 25-40 (Part 1: the Code of Ethics for Engineers), plus past-paper questions on competence, responsibility and cases

This section covers the first part of the engineering ethics lecture. It starts with role responsibilities, moves to the structure of the engineering code and its fundamental canons, works through the foreign-bolts and Larom cases, and ends with the point that ethics is practised, not memorised. The full canon lists, the functions of codes and their limitations are in section t11. The law versus morality grid (slides 41-43) is in section t06. Why ethics belongs in engineering at all is in section t03.

Role responsibilities (slides 26-27)

Ethics applies to your life in more than one way. The lecturer makes the distinction through **role responsibilities**. A role responsibility is a responsibility that attaches to us **in virtue of a role that we have**. Nobody has just one role. At any moment you may be an engineering student, a friend, a citizen and an employee, and each role carries its own duties.

Slide 27 gives five examples. Reproduce them as pairs of role and duty:

Role	Role responsibility
Friend	Look out for the interests of your friend.
Athlete	Play your sport in a professional manner.
Employee	Perform the duties of your job.
Parent	Look after your children and their interests.
Citizen	Follow the laws of the country in which you live.

The point of the table is that the engineer is one more role in the list. Being an engineer puts duties on you that a non-engineer does not have. The engineering code of ethics is the written statement of those duties. Compare this with the earlier argument in section t03: ethics comes into engineering through social responsibility, and social responsibility then becomes professional responsibility for each individual.

> **Key point.** A role responsibility is a responsibility that attaches to a person because of a role they have. Different roles give different duties, and one person holds several roles at once. Roles can conflict, which is where dilemmas come from (see t06).

MARKER WORDS: ROLE RESPONSIBILITIES

Say it like this:

> Role responsibilities are **responsibilities that attach to us in virtue of a role** that we have, such as friend, employee, parent, citizen or engineer.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
role responsibilities	explaining why an engineer has duties an ordinary person does not	SHOULD
in virtue of a role	defining the term	SHOULD
professional responsibility	linking the engineer's role to ethics	MUST
social responsibility	explaining why engineering needs ethics	MUST
duties and obligations	describing what a role puts on an individual	SHOULD

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define role responsibilities.
Responsibilities that attach to us in virtue of a role that we have. Each of us plays different roles in life.

The role responsibility of a friend, athlete, employee, parent and citizen.
Friend: look out for your friend's interests. Athlete: play your sport in a professional manner. Employee: perform the duties of your job. Parent: look after your children and their interests. Citizen: follow the laws of the country in which you live.

Ethics across the fields of engineering (slide 28)

There are many fields of engineering: civil, mechanical, electrical, software, industrial and others. The skills differ, but **the ethical issues arise across all of them**. A civil engineer who skimps on a bridge support, an electrical engineer who ignores a fault, and a software engineer who ships code known to be unsafe are all facing the same kind of choice: safety against cost, honesty against pressure.

Because of this, **the code of ethics for engineers pertains to engineers of all kinds**. One code serves every branch. That is why the canons are written in general words (public safety, competence, truthfulness) and not

in terms of any one technology.

Why follow the code? (slide 29, clicker)

The slide asks: engineers should follow their professional code of ethics because...

1. A. The public will trust engineers more if they know engineers have a code of ethics.
2. B. It helps them avoid legal problems, such as getting sued.
3. C. It provides a clear definition of what the public has a right to expect from responsible engineers.
4. D. It raises the image of the profession and hence gets engineers more pay.

The slide does not print the answer. `beyond the slides` The best answer is **C**. A and D describe side effects (trust, image, pay) and B is self-protection. Those are selfish or incidental reasons. C is the real purpose: the code sets out, in clear terms, what the public may rightly expect of a responsible engineer. This matches the duty-based reasoning of the rest of the lecture, where engineers act because it is right, not because it pays. The functions of a code are listed in full in t11.

The three components of the code (slide 30)

The Engineering Code of Ethics has **three components**. Learn them as a trio, with the verb that the slide attaches to each:

Component	What it does
The Fundamental Canons	Articulate the basic components of ethical engineering.
The Rules of Practice	Clarify and specify in detail the fundamental canons of ethics in engineering.
Professional Obligations	Elaborate the obligations that engineers have.

Think of it as three levels. The canons are the short, broad statements. The rules of practice take each canon and spell out what it means in specific situations. The professional obligations describe what engineers owe (to the public, employers, clients and the profession; see t09).

The NSPE fundamental canons (slide 31)

The slide gives the National Society of Professional Engineers canons. They all begin with the same stem: *Engineers, in the fulfillment of their professional duties, shall:*

1. Hold paramount the safety, health, and welfare of the public.
2. Perform services only in areas of their competence.
3. Issue public statements only in an objective and truthful manner.
4. Act for each employer or client as faithful agents or trustees.
5. Avoid deceptive acts.
6. Conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

> **Watch out.** This is the six-canon NSPE list from the slide. Exam questions ask for **seven** canons from the Uganda (UIPE) or IEEE/ABET-style codes, and the wording differs slightly (for example, canon 1 in the UIPE version adds sustainable development). Use the exact seven-canon lists in section t11 when the question says "list seven".

Canon 1 is the top priority. `beyond the slides` The word *paramount* means this canon overrides the others when they conflict. Public safety comes before loyalty to an employer or client.

MARKER WORDS: THE CODE AND ITS CANONS

Say it like this:

> The Engineering Code of Ethics has three components: **the Fundamental Canons**, which articulate the basic components of ethical engineering; **the Rules of Practice**, which clarify and specify the canons in detail; and **Professional Obligations**, which elaborate the obligations engineers have.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
hold paramount the safety, health and welfare of the public	giving the first canon or ranking duties	**MUST**
areas of their competence	stating the principle of individual competence	**MUST**
objective and truthful manner	discussing public statements or reports	**MUST**
faithful agents or trustees	describing duty to an employer or client	**MUST**
avoid deceptive acts	any case involving hiding or misreporting facts	**MUST**
enhance the honor, reputation and usefulness of the profession	the closing canon	**SHOULD**
Rules of Practice and **Professional Obligations**	naming the components of the code	**SHOULD**

▶ **Recall drill: the three components, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

The three components of the Engineering Code of Ethics.
The Fundamental Canons; the Rules of Practice; Professional Obligations.

What are the Fundamental Canons?
They articulate the basic components of ethical engineering.

What are the Rules of Practice?
They clarify and specify in detail the fundamental canons of ethics in engineering.

What are Professional Obligations?
They elaborate the obligations that engineers have.

The six NSPE fundamental canons.
Hold paramount the safety, health and welfare of the public; perform services only in areas of their competence; issue public statements only in an objective and truthful manner; act for each employer or client as faithful agents or trustees; avoid deceptive acts; conduct themselves honorably, responsibly, ethically and lawfully so as to enhance the honor, reputation and usefulness of the profession.

The principle of individual competence

This principle is canon 2: **engineers shall perform services only in areas of their competence**. It appears as a one-mark or three-mark question in several papers. Competence means having the education, training and experience needed to do the work safely. The slides tie it to student life too: to be competent you must actually know what you claim to know, which is why cheating undermines it (see t05).

Q: **CAT 1 2015 Q2a · 1 mark (also 2015 exam Q1a, 3 marks)** **State the principle of individual competence of an engineer.**

Try it yourself: the foreign bolts (slides 32-34)

The case. You are supervising a product whose specifications say that only U.S.-made parts may be used. Late in the project you discover that a sub-contractor has supplied a part with foreign-made bolts. They are not very noticeable and would function identically to U.S.-made bolts. Your customer urgently needs the finished product. What should you do?

The four options (clicker).

1. A. Say nothing and deliver the product with the foreign bolts because the customer will not notice.
2. B. Find some roughly equivalent violation of the contract or specs for which the customer is responsible, and tell them you will ignore their violation if they ignore yours.
3. C. Tell the customer about the problem, and let them decide what you should do next.

4. D. Find loopholes in the original specifications so that your company has not legally violated the specs.

The answer on the slide: C. Telling the customer is correct **because it lets the customer decide what is in their best interest given new information**. The slide admits this may be tough: your job may be on the line and your company's reputation may be at stake. It then names the two canons that justify the answer: **avoid deceptive acts** and **act for each employer or client as faithful agents or trustees**.

Why the others fail, in marker language: A is a deceptive act by silence. B is a bargain that treats honesty as a trade and drags the customer into a cover-up. D is deception by technicality; it keeps the letter of the contract while betraying its purpose. Only C treats the customer as a person entitled to the facts. Note that the bolts function identically, so this case is not about safety. It is about honesty and the duty to a client, which is why it is a good example of canons 4 and 5 working without canon 1.

MARKER WORDS: THE FOREIGN BOLTS CASE

Say it like this:

> I should **tell the customer about the problem and let them decide**, because that lets them choose what is in their best interest given new information. This follows the canons **avoid deceptive acts** and **act as faithful agents or trustees**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
faithful agents or trustees	justifying disclosure to a client	**MUST**
avoid deceptive acts	rejecting silence or loopholes	**MUST**
new information	explaining why the customer should decide	**SHOULD**
reputation of the company / **job on the line**	admitting the cost of doing the right thing	**SHOULD**

▶ **Recall drill: write the answer from memory, then click each blank to check**

Important notes about the code (slide 35)

The slide makes four short points. Each is a possible short-answer or discussion statement.

- **It is not a legally binding document.** The code is not a statute. Breaking it is not automatically a crime, although a professional body may discipline a member (t11 covers enforcement as a limitation). The law versus ethics contrast is in t06.
- **It is not something we want (or need) engineers to memorise.** Reciting canons proves nothing.

- **It is something we want engineers to understand and be able to live by.** The goal is conduct.
- **In the beginning, knowing the code is a guide to understanding how to apply it.** A beginner needs the canons as a map, so you should know them until applying them becomes natural.

> **Watch out.** The slide says memorising is not needed for real practice. It does not say you can skip learning the canons for the CAT. The exam asks you to list them, so learn them.

The Larom case (slides 36-38)

Setting. You were hired at Larom because of your promising research on catalysts as a student at SJSU. Your supervisor, Alex Smith, announces that your unit must make a recommendation within the next two days on which catalyst should be used in processing a major product.

The evidence. The overwhelming consensus of the engineers in your unit, based on many years of experience, is that catalyst A is best for the job. Your own research gives **preliminary evidence** that catalyst B might be **more reliable, more efficient and considerably less costly**.

The pressure. You ask whether the recommendation can be delayed a month to see if firmer evidence can be found. Alex asks you instead to write up the report **leaving out the preliminary data about catalyst B**. He says: "We've already taken too much time on this project. We have to be decisive, and we have to look decisive. Besides, we've had a lot of experience in this area."

Your position. You have no desire to challenge your colleagues. You do not necessarily disagree with them about which catalyst is best. But you wish you had been given more time to work on catalyst B, and you feel uncomfortable about leaving the preliminary data out of the report.

The task (slide 38). Discuss with a group: identify the issues, compare courses of action, identify the best course of action. One person reports out.

Worked analysis

`beyond the slides` The slides leave the answer open. Here is a model analysis in the template order (full template in t07).

Issues.

- **Honesty and deception.** Omitting data that points the other way gives the readers a false picture. Canon: *avoid deceptive acts*; *issue public statements only in an objective and truthful manner* (applied to an internal report).
- **Duty to the employer.** The company may spend money on a worse, costlier catalyst. Canon: *faithful agents or trustees*. Real loyalty means giving the employer the whole evidence.
- **Competence and uncertainty.** The data is only preliminary, so it must not be overstated either. Being honest includes saying how firm the evidence is.
- **Pressure and conformity.** Appearing decisive and relying on the experience of senior colleagues is not evidence.
- **Public safety and welfare.** The case says catalyst B may be more reliable, which can matter for a process affecting a major product.

Courses of action.

1. Write the report as ordered and leave the data out. Easy and obedient, but deceptive, and it breaks the canons above.
2. Include the data, clearly labelled *preliminary*, and say it needs more testing. Honest and does not overclaim.
3. Go to Alex first, explain the concern, and ask again for a short extension or for a line on catalyst B to be added.
4. If Alex refuses, take it to a higher manager or the ethics channel, without making accusations.
5. Refuse to write the report at all. Strong, but it gives up your chance to influence the outcome.

Best course. Combine 3 and 2, with 4 as a fallback. Talk to Alex first, propose adding the preliminary catalyst B results with an honest caveat about their limits, and keep the recommendation for A if that is what the evidence supports. Do not leave the data out. This respects Alex's need for a decision, keeps you honest, and gives the employer the facts. You do not have to say A is wrong. You only have to avoid hiding what you found.

MARKER WORDS: THE LAROM CASE

Say it like this:

> I should not leave out the preliminary data. I should **discuss it with my supervisor**, include it **clearly labelled as preliminary**, and if he refuses, **escalate**, because omission is a **deceptive act** and I must act as a **faithful agent** of my employer.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
preliminary data / preliminary evidence	describing the research and its limits	**SHOULD**
avoid deceptive acts	rejecting the omission	**MUST**
faithful agents or trustees	explaining why full information serves the employer	**MUST**
identify the issues, **compare courses of action**, **best course of action**	structuring any case answer	**SHOULD**
courage and **integrity**	explaining what it takes to resist the pressure	**SHOULD**

▶ **Recall drill: write the recommendation from memory, then click each blank to check**

The Atlanta Olympic stadium light tower

This is a classic "canon 1" case about public safety and communication.

Q: **2016 Q3C · 9 marks** **In March 1995, the structural engineer for the Olympic Stadium in Atlanta, Georgia, discovered an error in his design of a steel light tower to be built as part of the stadium. He told the stadium's architect, sending along a sketch to repair the error. Supposing that work on the light tower was a month away, he treated the redesign as a routine change instead of an emergency order. In fact, construction was ahead of schedule and work on the light tower began ten days after he informed the architect of the error and the need for repair. The repairs had not been carried out by the time an ironworker fell fifty feet to his death when the canopy holding the lights broke. (i) What ethical issues, if any, does this incident raise? (ii) Is there anything the engineer should have done differently? Why? [reconstructed from garbled scan]**

Two engineers, sealed bids

Q: **2015 Q3d · 7 marks (also 2011/12 paper)** **Two engineers submitted sealed bids to a prospective client for an engineering project. The client informed Engineer A of the amount that Engineer B had bid and invited Engineer A to resubmit a lower bid. Analyze this dilemma from the perspective of the client and Engineer A.**

Personal versus organizational responsibility

beyond the slides Set 2 gives no definition for this pair, but it is a repeat question. Build the answer from the role-responsibility idea and general knowledge.

Q: **2015 Q1c · 8 marks (also 2011/12 paper)** **Distinguish between personal responsibility and organizational responsibility as far as morality is concerned.**

MARKER WORDS: COMPETENCE, BIDS AND RESPONSIBILITY

Say it like this:

> **Personal responsibility** is the moral accountability of an individual for their own actions, while **organizational responsibility** is the moral accountability of the company as a whole for its policies, structures and culture. The first canon says to **hold paramount the safety, health and welfare of the public**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
hold paramount the safety, health and welfare of the public	the Atlanta tower case or any safety failure	**MUST**
areas of their competence	the competence principle	**MUST**
avoid deceptive acts	the sealed-bid dilemma	**MUST**
faithful agents or trustees	weighing duty to the client	**MUST**
professional responsibility	owning one's own work and follow-up	**MUST**
personal / organizational responsibility	distinguishing individual from corporate accountability	**MUST**

▶ **Recall drill: write the competence principle from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

State the principle of individual competence of an engineer.
Engineers shall perform services only in areas of their competence.

Four points about the code of ethics (slide 35).
It is not a legally binding document; we do not want (or need) engineers to memorize it; we want engineers to understand it and live by it; in the beginning, knowing the code is a guide to understanding how to apply it.

Ethics takes practice: knowledge versus behaviour (slide 39)

Unlike a robot, nobody can simply program you to be an ethical engineer who follows the codes. The slide makes three points.

- No one can program you to follow the code.
- **It is possible to know the codes of ethics yet fail to follow them.** Knowledge is not behaviour.
- **Ethical behaviour is about practice and virtue.** It is about going beyond the codes and practising behaviour that leads to an ethical life.

This is the same idea as the note that the code need not be memorised: a person can recite all the canons and still lie in a report. What counts is the habit of acting well, and habits are built by practice.

The example of integrity (slide 40)

The slide uses an analogy from structures.

- A building has **structural integrity** when it is designed so that it responds appropriately to the stresses and loads it is designed to carry.
- A building can have poor integrity or good integrity, and **so can a person**.
- A person has **integrity** when he or she can **follow the codes he or she is supposed to follow under the stresses and loads of his or her role**.

The loads on a person are things like the pressure to look decisive (Larom), the pressure of an urgent customer (bolts), or the fear of losing a job. A bridge that holds only on a calm day has no integrity; a person who follows the code only when it costs nothing has none either. The closing slide of the part (slide 58, which you will meet again in t05) says being ethical needs **courage** to do the right thing the situation calls for, and **the integrity to withstand the pressures** that push you in the wrong direction.

> **Key point.** Integrity is the ability to follow the codes under pressure. Knowing the code is not enough. Courage and integrity are the practical side of ethics.

MARKER WORDS: KNOWLEDGE, PRACTICE AND INTEGRITY

Say it like this:

> A person has **integrity** when he or she can **follow the codes under the stresses and loads of his or her role**, just as a building has structural integrity when it responds appropriately to the loads it is designed for.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
integrity	explaining what lets a person resist pressure	MUST
stresses and loads	using the building analogy	SHOULD
knowledge versus behaviour	explaining why knowing the code is not enough	SHOULD
practice and virtue	describing ethical behaviour	SHOULD
not legally binding	describing the status of the code	SHOULD
understand and live by	saying what we want from engineers	SHOULD

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

When does a person have integrity (slide 40)?
When he or she can follow the codes he or she is supposed to follow under the stresses and loads of his or her role.

When does a building have structural integrity?
When it is designed in a way that appropriately responds to the stresses and loads that it is designed to act under.

What two things does being ethical require (slide 58)?
Courage to do the right thing the situation calls for, and the integrity to withstand the pressures that push you in the wrong direction.

Academic integrity and ethical conduct of research

~25 min to read | Medium priority | 5 questions

Source: Set 2, slides 44-58; Set 3, slides 2-3. Research-misconduct material marked "beyond the slides" is added from general knowledge.

Students have a code of ethics too

Slide 44 opens "Part 2: Practicing ethics as an engineering student". The lecturer moves from the engineer's code of ethics to the student's own code. The idea is that you practise ethics now, as a student, so you are ready to practise it as an engineer.

The code used as the example is Senate Policy S07-2, the Academic Integrity Policy of San Jose State University (SJSU). It requires that each student:

1. Knows the rules that preserve academic integrity and abides by them at all times. This includes learning and abiding by the rules for specific classes, exams and course assignments.
2. Knows the consequences of violating the Academic Integrity Policy.
3. Knows the appeal rights, and the procedures to be followed in the event of an appeal.
4. Fosters academic integrity among peers.

Notice item 4. A student is not only asked to be honest personally. The student must also help others to be honest. This mirrors the engineering canons, where an engineer must also act against colleagues' unethical conduct.

The S07-2 preamble

The preamble gives the reason for the policy. In the slide's words:

- The university emphasizes *responsible citizenship* and an awareness of the *ethical choices inherent in human development*.
- *Academic honesty and fairness* foster ethical standards for all who depend upon the integrity of the university, its courses and its degrees.
- University degrees are *compromised* and the public is *defrauded* if faculty members or students, knowingly or unwittingly, allow dishonest acts to be rewarded academically.

Two points to carry into an essay. First, the harm of cheating reaches beyond the cheater and the class: it reaches everyone who relies on the degree (employers, patients, the public). Second, even *unwittingly* allowing dishonesty to be rewarded is a failure. A lecturer or classmate who looks away shares in the damage.

Plagiarism and cheating: why competence is the issue

Slide 47 links cheating to the engineering codes of ethics. Its argument runs in four steps:

1. Many components go into being a good engineering student.
2. One of the most important, as reflected in the codes of ethics for engineers, is to be *competent* in your field. (Link to the principle of individual competence in the code of ethics section.)
3. To be competent, you must actually *know what you claim to know*.

4. Proving to others that you know what you are supposed to know requires *certification through a degree*.

So a degree is a public claim of competence. Cheating makes the claim false. That is why cheating is not a small school matter but an early breach of the same duty of competence you will owe as an engineer.

What students say: the statistics

Slide 48 gives three figures about how common cheating is:

- **70%** of American high school seniors admit to cheating on at least one test.
- **95%** of the students who said they cheated were *never caught*.
- An average of **75%** of college students report cheating sometime during their college career.

Use them to make the point that cheating is widespread and rarely punished, so the pressure to join in is real. That is exactly why the course ends the topic on courage and integrity (below). The figures are American; do not claim them for Uganda.

FLASHCARDS: CLICK TO FLIP

The four things S07-2 requires of each student.
1. Know the rules that preserve academic integrity and abide by them at all times. 2. Know the consequences of violating the policy. 3. Know the appeal rights and procedures. 4. Foster academic integrity among peers.

The three cheating statistics.
70% of American high school seniors admit to cheating on at least one test; 95% of those who cheated were never caught; an average of 75% of college students report cheating sometime in college.

Cheating defined, and the five SJSU types

Slide 49 gives the definition. At SJSU, **cheating is the act of obtaining or attempting to obtain credit for academic work through the use of any dishonest, deceptive, or fraudulent means.** Note "attempting": you do not have to succeed. Note "credit for academic work": the wrong is claiming a reward you did not earn.

SJSU defines five basic types of cheating. Each has its own slide, retold here.

Type	What the slide says	Example
1. Copying	Copying someone's work on a homework assignment, exam or paper. Submitting someone's work as your own.	Copying a classmate's lab report.
2. Multiple submissions	Submitting your own work from one class to another class, or submitting one piece of work to two distinct classes. A paper for one class is not a paper for another class.	Handing in last semester's essay again for a new course.
3. Unauthorized sources	Using sources you are not allowed to use, as deemed by the instructor or the university as a whole.	Solution manuals; a friend's text message with an exam answer.
4. Altering grades	Altering your grade in any way.	Changing a C to a B+ on a marked paper.
5. Surrogate cheating	Someone else does your homework, takes an exam for you, or writes your paper. Doing someone's work for them is also cheating.	Paying someone to sit your exam or write your project.

Watch out. Plagiarism is not on the slide's list of five as a separate heading. It is a form of *copying* (submitting someone's work as your own). If asked about plagiarism, name it as copying and describe it that way.

Q: **Slides 49-56 · exam-style · 5 marks** **Name and describe five types of cheating that a student can commit.**

Cheating versus teamwork, and free-riding

Slide 51 stops students from drawing the wrong conclusion. Working on a team for an assigned project is *not* cheating. Teamwork is important in engineering. But failing to do your assigned task on a team project *is* a form of cheating. It is called **free-riding**: benefiting from the work of others without doing any work of your own.

The slide's reason why free-riding is wrong is a universalization argument (compare Kant in the ethics foundations section): if everyone did it, nothing would get done. Say this when you explain it, because it is the justification the slide gives.

Why cheating is wrong

The slides give two separate harms.

1. It harms honest students (slide 50)

Cheating undermines the work of fellow students who are honest. When you cheat, all the students who did not cheat are penalized. They end up with lower grades, and as a consequence of lower grades they lose out on scholarships and recommendations. This works when marks are relative or ranked: a dishonest high score pushes honest scores down.

2. It undermines the credibility of the university and its degrees (slide 57)

If too many people cheat at the university, its degrees will not certify that its students are competent. So by cheating you not only hurt yourself, you also hurt others. This returns to the preamble: degrees are compromised and the public is defrauded.

3. Beyond the slides

beyond the slides A third argument is about the cheater. A cheater who graduates without the knowledge may later design something unsafe, which links to the engineer's duty to the public. A fourth is that cheating breaks the trust and honesty that make any profession work.

> **Key point.** To "explain why cheating is wrong", give at least these: it penalizes honest students (lower grades, lost scholarships and recommendations); it undermines the credibility of the university and its degrees; it hurts the cheater as well as others; and, in a team, free-riding fails the "if everyone did it" test.

Ethics: courage and integrity

The last slide (58) says that, as the course will show more and more, being ethical requires two things:

- **Courage** to do the right thing the situation calls for.
- The **integrity** to withstand the pressures that push you in the wrong direction.

Knowing the rules is not enough. Given the statistics (95% never caught), the temptation is strong and the honest student needs these two qualities. They return later in whistleblowing and in cases such as the engineering code cases.

MARKER WORDS: STUDENT CODE, CHEATING AND WHY IT IS WRONG

Say it like this:

> Cheating is the act of **obtaining or attempting to obtain credit for academic work** through any **dishonest, deceptive, or fraudulent means**. It is wrong because it **undermines the credibility** of the university and its degrees and **penalizes honest students**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
competent / competence	linking cheating to the engineer's duty; a degree certifies competence	**MUST**
credibility of the university and its degrees	explaining why cheating hurts others	**SHOULD**
academic integrity	naming what S07-2 protects	**SHOULD**
free-riding	separating teamwork from cheating	**SHOULD**
surrogate cheating	listing the five types	**SHOULD**
unauthorized sources	listing the five types	**SHOULD**
scholarships and recommendations	showing how honest students lose	**SHOULD**
courage and integrity	saying what ethical action requires	**SHOULD**

▶ **Recall drill: write the definition from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

Define cheating (SJSU).
The act of obtaining or attempting to obtain credit for academic work through the use of any dishonest, deceptive, or fraudulent means.

The five SJSU types of cheating.
Copying; multiple submissions; unauthorized sources; altering grades; surrogate cheating.

Define free-riding.
Benefiting from the work of others without doing any work of your own.

Define surrogate cheating.
Someone else either does your homework, takes an exam for you, or writes your paper.

What does being ethical require (slide 58)?
Courage to do the right thing the situation calls for, and the integrity to withstand the pressures that push you in the wrong direction.

The hiring exercise and real cheating scandals

Set 3 opens with a review exercise from Lecture 1. You are an employer at a large multinational software firm. You advertise on Monster.com for software engineers to design a new product. Many applications arrive, but two stand out because of their internships and their communication skills in interviews.

- **Applicant 1** has almost all A grades, but comes from a university where several cheating scandals have been reported in the newspapers.
- **Applicant 2** has mostly B grades, but comes from a university well known for strictly enforcing integrity.

The question: who would you hire? The slide does not give the answer. It is designed to show that a degree is only as good as the integrity behind it.

Q: `Set 3 slide 2 · discussion` **You must choose between an applicant with almost all A grades from a university with reported cheating scandals, and an applicant with mostly B grades from a university known for strictly enforcing integrity. Who would you hire, and why?**

MARKER WORDS: THE HIRING EXERCISE

Say it like this:

> Where cheating is tolerated, grades no longer **certify that the student is competent**, so I weigh the **credibility of the degree** and the applicant's **integrity** above the raw grade.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
credibility of the degree	justifying the second applicant	**SHOULD**
competent	saying what a degree should certify	**SHOULD**
integrity	saying what the employer needs	**SHOULD**

Slide 3, headed "This really happens", lists real reports to show that these scandals are not hypothetical:

- "Insider: UNC tolerated cheating", Charlotte Observer, 18 November 2012.
- U.S. News strips George Washington University of its ranking for cheating, 15 November 2012.
- Cheating scandal rocks Harvard University, NECN.com, 31 August 2012.
- An assistant registrar accused of changing hundreds of student grades at Southern University, 2003.
- Diablo Valley College students accused of trading sex for grade changes, 2007.

Match these to the types above. The registrar and the Diablo Valley cases are *altering grades*, committed by an insider. The ranking and tolerance cases show a *university's credibility* being lost, with consequences for every student who holds its degree. Note that the cheating here is not limited to students: staff and institutions cheat too. The Jenny case (cheating allowed at job exams) is worked in the dilemmas section.

FLASHCARDS: CLICK TO FLIP

The five real scandals on the "This really happens" slide.
UNC tolerated cheating (2012); George Washington University stripped of U.S. News ranking (2012); Harvard cheating scandal (2012); Southern University assistant registrar changed hundreds of grades (2003); Diablo Valley College students traded sex for grade changes (2007).

Ethical conduct of research

beyond the slides The slides teach academic integrity through cheating and plagiarism. Exams ask a research version: "identify and explain three practices that violate ethical conduct of research". The same values apply (honesty, credit to others, accuracy) but to research and reports such as a final-year project. Practices to choose from:

1. **Plagiarism**: presenting other people's words, data or ideas as your own, without quotation marks and citation. This is the slides' *copying* applied to a report.
2. **Fabrication**: inventing data or results that were never collected.
3. **Falsification**: changing, omitting or manipulating data, equipment or processes so the research does not truly represent what was found.
4. **Selective use of data**: keeping only results that support your hypothesis and hiding the rest.
5. **Authorship abuse**: leaving out people who contributed, listing people who did not (gift authorship), or claiming credit for a student's or colleague's work.
6. **Multiple or duplicate submission**: publishing or submitting the same work twice as new (the slides' multiple submissions).
7. **Breach of participant rights**: collecting data without informed consent or breaking confidentiality.

The first two or three are the safest to choose, since they connect directly to the slides' definition of cheating as *dishonest, deceptive, or fraudulent means*.

Q: **Past papers 2011-2015 · 6 marks** **Identify and explain three (3) practices that violate ethical conduct of research.**

The Mildred case

This case appears in the 2015, 2013/14 and 2011/12 papers with identical wording. The questions on it are: (i)

> EXAM WORDING
> Describe Mildred's ethical dilemma

(**2 marks**); (ii)

> EXAM WORDING
> Is Mildred's final project report a product of ethical conduct of research? Justify

(**3 marks**). Retold:

Mildred is a fourth-year undergraduate student in the Faculty of Technology (CEDAT) preparing her Final Year Project Report. Her project needs a survey among three mobile telephone service providers in Uganda. A few weeks before her final examinations she realizes she cannot get data from two of the three companies. She draws similar statistics from a report by the Uganda Communications Commission to fill the gap. To get her work ready for her supervisor, she then puts whole sentences and paragraphs, word for word, from several published papers and the internet into her report. She uses no quotation marks or footnotes to show the other authors consulted.

Reading the case

- **Pressure:** time (only a few weeks left), missing data, and a supervisor waiting.
- **Substituted data:** statistics from the UCC report were used in place of data she was meant to collect. Using them is not wrong by itself. Presenting them as results of her own survey, or not saying that two of the three companies were not surveyed, would be misrepresentation. The case does not say whether she disclosed it.
- **Plagiarism:** verbatim sentences and paragraphs with no quotation marks or footnotes. This is copying (slide 52): submitting someone's work as your own.
- **Honest options she had:** tell her supervisor early, narrow the project to the one company she could survey, state the limitation, cite the UCC report openly, and quote and reference every source.

Q: **Past papers 2015 / 2013-14 / 2011-12 · 2 marks** **(d)(i) Describe Mildred's ethical dilemma.**

Q: **Past papers 2015 / 2013-14 / 2011-12 · 3 marks** **(d)(ii) Is Mildred's final project report a product of ethical conduct of research? Justify.**

MARKER WORDS: ETHICAL CONDUCT OF RESEARCH AND PLAGIARISM

Say it like this:

> Plagiarism is **submitting someone else's work as your own** without **quotation marks or citation**. It violates ethical conduct of research because it is **dishonest**, so the report is not a product of ethical research.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
plagiarism	naming the main violation in Mildred's report	**MUST**
without quotation marks or footnotes	describing exactly what Mildred did wrong	**MUST**
dishonest, deceptive, or fraudulent means	justifying why it counts as cheating	**SHOULD**
submitting someone's work as your own	defining copying	**SHOULD**
credibility	concluding the harm to her report and degree	**MUST**
fabrication / falsification	listing research violations (beyond the slides)	**SHOULD**

▶ **Recall drill: write the sentence from memory, then click each blank to check**

FLASHCARDS: CLICK TO FLIP

What is copying (cheating type)?
Copying someone's work on a homework assignment, exam or paper; submitting someone's work as your own.

What are multiple submissions?
Submitting your own work from one class to another class, or one piece of work to two distinct classes.

Three research violations to cite in an exam answer (beyond the slides).
Plagiarism; fabrication of data; falsification or selective use of data. Others: authorship abuse, duplicate submission.